GRE Data Interpretation Drill Sheet

Graph Taxonomy, Axis Scale Audits, Percentage Change Formulas & Estimation Shortcuts

1. The 5 Core GRE Data Interpretation Formats

Data Interpretation (DI) accounts for 3 to 4 questions per Quant section, typically presented in a single comprehensive graph set.

Graph / Chart Archetype	Visual Representation & Format	Primary Analytical Focus	Common Visual Trap
Bar Graphs (Segmented/Stacked)	Vertical or horizontal bars showing discrete category totals	Reading individual segments; calculating component percentages	Confusing total bar height with individual segment length
Line Graphs (Time Series)	Continuous line tracking longitudinal trends over months/years	Calculating slope, rates of change, and finding peaks/troughs	Misidentifying steep slope with highest absolute value increase
Pie Charts (Proportional)	Circular charts showing percentage or degree breakdown of a whole	Converting percentages to raw numbers; degree conversions (360 deg = 100%)	Assuming equal slice sizes mean equal numbers when base totals differ!
Scatterplots & Trendlines	Coordinate plane plotting paired data points with line of best fit	Identifying positive/negative correlation, outliers, and clusters	Projecting trends beyond the bounded axis range
Complex Multi-Column Tables	Spreadsheet-style rows and columns with numerical data	Cross-referencing row and column intersections quickly	Reading the wrong row due to eye tracking errors

Section Budget: Spend 1.5 to 2 minutes reading the initial graph setup; budget 1m30s per associated question.

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2. High-Yield Data Interpretation Mathematical Formulas

Memorize these foundational calculation formulas that appear in virtually every GRE DI set.

Analytical Metric	Standard Mathematical Formula	Mental Math Shortcut / Trick	Trap to Avoid
Percentage Change (%)	((Final Value - Initial Value) / Initial Value) * 100	Calculate the difference first; estimate fraction before dividing	Dividing by the Final Value instead of the Initial baseline!
Percentage Of vs More Than	A is x% of B: A = (x/100)*B | A is x% more than B: A = (1 + x/100)*B	125% of B = 1.25*B; 25% more than B = 1.25*B	Confusing "what percent of" with "what percent greater than"
Pie Chart Degree to %	Percentage = (Degrees / 360) * 100	Every 36 degrees represents exactly 10% of the total pie	Treating degrees as percentages directly without dividing by 360
Weighted Average of Groups	Weighted Mean = (n1*X1 + n2*X2) / (n1 + n2)	The average always pulls closer to the group with larger size	Taking the simple arithmetic mean when group sizes are unequal!

On-Screen Calculator Tip: The GRE provides an on-screen calculator; use it for 4-digit division, not simple arithmetic.

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3. The 4-Step Systematic Data Interpretation Attack Plan

Execute this exact sequence on every DI question to eliminate careless calculation errors.

- Step 1: Read the Title, Legend, and Axis Labels FIRST. Note units (thousands vs millions, dollars vs euros) before reading questions.
- Step 2: Check the Answer Choices for Spread. If answer choices differ by > 15%, round aggressively and estimate rather than computing exact decimals.
- Step 3: Track the Denominator Base. In percentage questions, always ask: "Percent of WHAT?" The denominator is the most common trap.
- Step 4: Check Reasonableness. Does an answer indicating 850% growth make sense based on the visual slope? Sanity-check every answer.